

Parity Statistics of Nested Floor Powers

Conditional Descent Results for the Juggler Map

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Abstract

The Juggler map applies the integer part of the square root at even positive integers and of the three-halves power at odd positive integers. Its finite itineraries lead to joint parity questions for compositions of floor powers. We give a self-contained conditional framework relating these parity statistics to descent below the initial value. Exact branch-indicator identities and a power envelope reduce finite descent certificates to explicit correlation estimates along prescribed formal chains. The classical second-derivative estimate gives an unconditional certificate class of natural density $3/4$. Under stated finite-chain correlation hypotheses, the certificate classes of lengths at most four and five have densities $13/16$ and $7/8$, respectively. If every fixed-depth odd-rooted itinerary has its fair-share density, the set of starts admitting some finite power-envelope certificate has density one; no uniformity in depth is needed for this implication. We also record exact floor-defect identities, a Fourier criterion sufficient for the correlation hypotheses, and a shift-averaged cancellation estimate. The nested correlation and deterministic kernel estimates remain unproved here. In particular, the larger certificate densities and the density-one conclusion are conditional, and none asserts universal arrival at 1.

Keywords: Juggler map; nested floor powers; parity correlations; discrepancy; stopping time; conditional descent.

1. Scope and relation to earlier work

For a positive integer n , define

$$J(n) = \begin{cases} \lfloor n^{1/2} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

This is the Juggler map recorded in OEIS A094683 [1]. The conjecture that every positive orbit reaches 1 remains open. The question studied here is more limited: which distribution estimates for finite itineraries would imply that many starting values eventually fall below their initial value?

Single-floor parity estimates follow from classical exponential-sum methods [2, 3]. Composing floors creates an additional difficulty. For $m = \lfloor n^{3/2} \rfloor$ and $\theta = \{n^{3/2}\}$,

$$m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4}).$$

The coefficient of the fractional part grows with n . A theorem for the smooth phase $n^{9/4}$ therefore does not by itself estimate the nested phase. Nor does a parity estimate over consecutive starting values automatically apply to a sparse set of orbit images.

The contribution of this note is an explicit reduction: the exact identities, finite certificate classification, sufficient analytic hypotheses, and conditional consequences are stated separately. The single-floor estimate and the probabilistic counting argument use standard methods. The latter is in the spirit of stopping-time arguments for the Collatz map, such as Terras [4]; no theorem about Collatz is transferred to the Juggler map.

This version replaces an earlier working draft that asserted nested parity equidistribution and kernel cancellation. Those analytic claims are not established by the arguments retained here. The powers $23/24$ and $95/96$ below are research targets, not proved error exponents. The former numerical threshold $3.6 \cdot 10^{13}$ is not a certified threshold for any nested cancellation theorem in this version. All proofs of conditional implications are given below; their analytic hypotheses remain explicit.

Throughout, $e(t) = \exp(2\pi it)$, $\{t\} = t - \lfloor t \rfloor$, and $\psi(t) = (-1)^{\lfloor t \rfloor}$. Natural density is measured among all positive integers. Thus a length- d odd-rooted word with count $2^{-d}N + o(N)$ has relative density $2^{-(d-1)}$ among odd starts. Constants in O and $\ll$ may depend on fixed words and explicit fixed parameters. They are not asserted to be uniform in depth.

2. Exact itineraries and the power envelope

Let $w = w_1 \cdots w_d$ be a word in $\{E, O\}$. The itinerary word $_d(n)$ records the parities of $n, J(n), \dots, J^{d-1}(n)$. It contains d letters and specifies the d operations that produce $J^d(n)$. Write $o(w)$ for the number of its odd letters.

Proposition 2.1 (power envelope). If w is realized at n , then

$$(J^d(n))^{2^d} \leq n^{3^{o(w)}}.$$

Consequently, if $n \geq 2$ and $3^{o(w)} < 2^d$, then $J^d(n) < n$.

Proof. At any positive integer x , the even operation satisfies $J(x)^2 \leq x$, and the odd operation satisfies $J(x)^2 \leq x^3$. Suppose a prefix of length r with o odd letters ends at x and obeys $x^{2^r} \leq n^{3^o}$. An even operation preserves the right-hand exponent after raising $J(x)^2 \leq x$ to 2^r ; an odd operation replaces it by 3^{o+1} . Induction starts at the empty prefix. If $3^{o(w)} < 2^d$ and $n > 1$, the bound is strictly less than n^{2^d} , proving descent. $\square$

A word satisfying $3^{o(w)} < 2^{|w|}$ is called *contracting*. A realized contracting prefix is a *power-envelope descent certificate*. Let

$$\mathcal{C}_d = \{n \geq 2 : \text{some contracting prefix of length at most } d \text{ is realized at } n\}.$$

Also set

$$\mathcal{C}_\infty = \bigcup_{d \geq 1} \mathcal{C}_d.$$

These sets refer to this particular sufficient certificate. Flooring can cause descent even when the envelope does not certify it. Therefore $\mathcal{C}_d$ need not equal the set of all starts that descend within d steps.

For an odd-rooted target word w , define its *formal chain* for every odd n , regardless of whether n realizes w :

$$x_1^w(n) = n, \quad \xi_t^w(n) = (x_t^w(n))^{p_t}, \quad x_{t+1}^w(n) = \lfloor \xi_t^w(n) \rfloor \quad (1 \leq t < d),$$

where $p_t = 3/2$ for $w_t = O$ and $p_t = 1/2$ for $w_t = E$. Set $s_t^w(n) = \psi(\xi_t^w(n))$, and put $\epsilon_t = 1$ for $w_{t+1} = E$, $\epsilon_t = -1$ for $w_{t+1} = O$.

Lemma 2.2 (parity and branch indicators). For real x , $\lfloor x \rfloor$ is odd exactly when $\{x/2\} \in [1/2, 1)$. For every odd n and odd-rooted word w ,

$$\mathbf{1}_{\text{word}_d(n)=w} = 2^{-(d-1)} \prod_{t=1}^{d-1} (1 + \epsilon_t s_t^w(n)). \quad (2.1)$$

Proof. Writing $x = \lfloor x \rfloor + \{x\}$ proves the first assertion, including the half-open endpoints. Each factor divided by two tests the parity of x_{t+1}^w . Until the first failed test, the formal chain agrees with the true orbit, by induction on t . A failed test makes the product zero and prevents realization of w . If all tests pass, the chains agree through the required letters and the product is one. The empty product covers $d = 1$. $\square$

The use of formal chains is essential: the product is defined on every odd input before any correlation or Fourier expansion is made.

3. An unconditional certificate density

Let $M(N) = \#\{n \leq N : n \text{ odd}\} = N/2 + O(1)$, and set

$$S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} \psi(n^{3/2}).$$

Theorem 3.1 (single-floor parity). One has

$$S_O(N) = O(N^{5/6}).$$

In particular,

$$\begin{aligned} \#\{n \leq N : \text{word}_2(n) = OO\} &= N/4 + O(N^{5/6}), \\ \#(\mathcal{C}_2 \cap [1, N]) &= 3N/4 + O(N^{5/6}). \end{aligned}$$

Proof. Write $n = 2r + 1$ and $g(r) = \frac{1}{2}(2r + 1)^{3/2}$. Then $g''(r) = \frac{3}{2}(2r + 1)^{-1/2}$. On a dyadic block with $r \asymp Q$, the classical second-derivative test [2] gives

$$\left| \sum_{r \asymp Q} e(hg(r)) \right| \ll h^{1/2} Q^{3/4} + h^{-1/2} Q^{1/4} \quad (h \geq 1).$$

The Erdős–Turán inequality [3, Chapter 2] bounds the unnormalized interval discrepancy by

$$\ll Q/H + \sum_{h=1}^H \frac{1}{h} \left| \sum_{r \asymp Q} e(hg(r)) \right| \ll Q/H + H^{1/2} Q^{3/4} + Q^{1/4}.$$

Taking $H = \lfloor Q^{1/6} \rfloor$ for sufficiently large Q gives $O(Q^{5/6})$. Lemma 2.2 identifies parity with membership in $[1/2, 1)$; summing the dyadic bounds proves the first assertion. Since $S_O = M - 2\#OO$, it also proves the count of OO .

The only minimal contracting words of length at most two are E and OE . The first class has count $\lfloor N/2 \rfloor$, and the second has count $M - \#OO = N/4 + O(N^{5/6})$. They are disjoint, and Proposition 2.1 proves the stated certificate count. $\square$

This argument estimates starting values in an interval. It makes no independence assertion about successive orbit parities.

4. Explicit correlation hypotheses

For a fixed odd-rooted word w of length d and nonempty $A \subseteq \{1, \dots, d-1\}$, define

$$R_{w,A}(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} \prod_{t \in A} s_t^w(n).$$

Hypothesis $H(w; \delta)$. For some fixed $0 < \delta < 1$, every nonempty A satisfies $R_{w,A}(N) = O_{w,A}(N^{1-\delta})$ as $N \rightarrow \infty$.

Hypothesis $H_0(w)$. The same correlations satisfy $R_{w,A}(N) = o(N)$, without a prescribed rate.

These are assertions about the formal chains, not assumptions that the true orbit is a sequence of independent coin tosses. Hypothesis $H(w; \delta)$ implies $H_0(w)$.

Proposition 4.1 (conditional word count). Under $H(w; \delta)$,

$$\#\{n \leq N : \text{word}_d(n) = w\} = 2^{-d}N + O_w(N^{1-\delta}). \quad (4.1)$$

Under $H_0(w)$, the error is $o(N)$.

Proof. Expand (2.1) and sum. The empty subset contributes $2^{-(d-1)}M(N) = 2^{-d}N + O(1)$. The other $2^{d-1} - 1$ terms are fixed signed multiples of $R_{w,A}$, and there are finitely many for fixed w . $\square$

For example, $H(OOEE; \delta)$ concerns the seven nonempty products of

$$\psi(n^{3/2}), \quad \psi(m^{3/2}), \quad \psi(v^{1/2}), \quad m = \lfloor n^{3/2} \rfloor, \quad v = \lfloor m^{3/2} \rfloor.$$

Proposition 4.1 then gives the count $N/16 + O(N^{1-\delta})$ needed for four-step certificates. Theorem 3.1 establishes only the first of these seven estimates; the others are not supplied here.

4.1 A sufficient Fourier criterion

For $\mathbf{k} = (k_1, \dots, k_{d-1}) \in \mathbb{Z}^{d-1}$, let

$$S_{w,\mathbf{k}}(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} e \left(\frac{1}{2} \sum_{t=1}^{d-1} k_t \xi_t^w(n) \right).$$

Proposition 4.2 (conditional Fourier-to-parity transfer). Fix $d \geq 2$, w , and $0 < \eta, \sigma < 1$. Suppose, uniformly over all nonzero integer vectors with $\|\mathbf{k}\|_\infty \leq \lfloor N^\eta \rfloor$, that

$$|S_{w,\mathbf{k}}(N)| \ll_w N^{1-\sigma}. \quad (4.2)$$

Then $H(w; \delta)$ holds for every $0 < \delta < \min(\eta, \sigma)$. The constants may depend on these fixed parameters.

Proof. Apply the multidimensional Erdős–Turán–Koksma inequality [3, Chapter 2, p. 116] to the points $\mathbf{y}_n = (\{\xi_1^w(n)/2\}, \dots, \{\xi_{d-1}^w(n)/2\})$. For any half-open axis-parallel box $B \subseteq [0, 1)^{d-1}$, its unnormalized discrepancy is

$$\begin{aligned} |\#\{n \leq N \text{ odd} : \mathbf{y}_n \in B\} - M(N)|B|| &\ll_d \frac{M(N)}{H} + \sum_{0 < \|\mathbf{k}\|_\infty \leq H} \frac{|S_{w,\mathbf{k}}(N)|}{r(\mathbf{k})}, \\ r(\mathbf{k}) &= \prod_{t=1}^{d-1} \max(1, |k_t|). \end{aligned}$$

The weight sum is $O_d((1 + \log H)^{d-1})$. With $H = \lfloor N^\eta \rfloor$, (4.2) gives box discrepancy $O_w(N^{1-\eta} + N^{1-\sigma}(1 + \log N)^{d-1})$. Each nonempty sign product is constant, with value 1 or -1 , on the

2^{d-1} boxes obtained by splitting each coordinate at $1/2$. Its integral over the unit cube is zero. Summing their discrepancies therefore bounds $R_{w,A}$; absorbing the fixed logarithmic power proves the assertion. $\square$

The criterion needs every indicated mixed mode, including vectors with zero coordinates and mixed signs. A bound for one undecorated kernel is not a substitute for (4.2). Qualitatively, cancellation $S_{w,\mathbf{k}}(N) = o(N)$ for each fixed nonzero $\mathbf{k}$ also implies $H_0(w)$: keep H fixed in the displayed inequality, let $N \rightarrow \infty$, and then let $H \rightarrow \infty$.

5. Conditional finite-depth descent densities

Lemma 5.1 (minimal certificates through length five). The contracting words of length at most five with no proper contracting prefix are exactly

$$E, \quad OE, \quad OOE, \quad OOOE, \quad OOEOE. \quad (5.1)$$

Proof. An E -rooted word already contracts at its first letter, and an OE -rooted word contracts at its second. Every remaining word begins with OO . At length three, even two odd letters give $3^2 > 2^3$, so no new certificate appears. At length four, contraction requires at most two odd letters, giving only $OOEE$. After excluding that prefix, the possible fourth-level prefixes are $OOEO$, $OOOE$, and $OOOO$. At length five, contraction requires at most three odd letters because $3^3 < 2^5 < 3^4$. The first two prefixes must therefore end with E , and the third cannot contract. This proves (5.1). $\square$

Theorem 5.2 (conditional four-step density). If $H(OOEE; \delta_4)$ holds for some $\delta_4 > 0$, then

$$\#(\mathcal{C}_4 \cap [1, N]) = \frac{13N}{16} + O(N^{1-\delta_*}), \quad \delta_* = \min(1/6, \delta_4).$$

The qualitative hypothesis $H_0(OOEE)$ instead suffices for natural density $13/16$.

Proof. Lemma 5.1 gives the disjoint decomposition into the prefix classes E , OE , and $OOEE$. Theorem 3.1 supplies the first two counts, and Proposition 4.1 supplies the third. Their densities add to $1/2 + 1/4 + 1/16 = 13/16$. $\square$

Theorem 5.3 (conditional five-step density). Suppose $H(w; \delta_w)$ holds for each

$$w \in \mathcal{W} = \{OOEE, OOOE, OOEOE\}, \quad \delta_w > 0.$$

Then

$$\#(\mathcal{C}_5 \cap [1, N]) = \frac{7N}{8} + O(N^{1-\delta_*}), \quad \delta_* = \min(1/6, \min_{w \in \mathcal{W}} \delta_w).$$

The three qualitative hypotheses $H_0(w)$ suffice for natural density $7/8$.

Proof. The two additional prefix classes in (5.1) are disjoint from each other and from the four-step classes. Each has density $1/32$ by Proposition 4.1. Thus the total is $13/16 + 1/32 + 1/32 = 7/8$. Lemma 5.1 proves that this list exhausts power-envelope certificates through length five. $\square$

The exact fractions and their analytic requirements can be read together:

Certificate length	Minimal prefixes added	Cumulative density	Status
1	E	$1/2$	Unconditional
2	OE	$3/4$	Theorem 3.1
4	$OOEE$	$13/16$	Conditional on $H_0(OOEE)$
5	$OOOE, OOEOE$	$7/8$	Conditional on the three hypotheses in Theorem 5.3

For the five-letter words, the four tested real images are, respectively, $(n^{3/2}, m^{3/2}, v^{3/2}, z^{1/2})$, with $z = \lfloor v^{3/2} \rfloor$, and $(n^{3/2}, m^{3/2}, v^{1/2}, u^{3/2})$, with $u = \lfloor v^{1/2} \rfloor$. Each hypothesis concerns all fifteen nonempty sign products of its four coordinates.

Separately, if $H_0(w)$ holds for every odd-rooted word of length four, all eight such classes have density $1/16$. This complete census is conditional as well; it is stronger than the single four-letter hypothesis used in Theorem 5.2.

6. The fixed-depth reduction to density-one certificates

Hypothesis FD. For every fixed $d \geq 1$ and every odd-rooted word w of length d ,

$$\#\{n \leq N : \text{word}_d(n) = w\} = 2^{-d}N + o_w(N).$$

No common error rate in d is assumed. The family $H_0(w)$ over all such words implies FD.

Theorem 6.1 (conditional density-one certificate theorem). Under FD, $\mathcal{C}_\infty$ has natural density one.

Proof. Put $p = \log 2 / \log 3 > 1/2$. A word with no contracting prefix through length d must in particular have at least pd odd letters at length d . All such words begin with O . For a uniformly chosen odd-rooted word of length d , the number of odd letters is distributed as $1 + B_{d-1}$, where B_{d-1} is a binomial random variable with parameters $(d-1, 1/2)$. This is an auxiliary count on words, not a stochastic model asserted for the Juggler orbit.

Choose $q = (p + 1/2)/2$, so $1/2 < q < p$. For all sufficiently large fixed d , $pd - 1 \geq q(d-1)$. For $t > 0$, the exponential Markov inequality and the binomial generating function give

$$\Pr(B_{d-1} \geq q(d-1)) \leq \left(\frac{1 + e^t}{2e^{qt}} \right)^{d-1}.$$

The logarithm of the expression in parentheses is zero at $t = 0$ and has derivative $1/2 - q < 0$ there. Hence some fixed $t > 0$ makes it a number $\vartheta < 1$.

For fixed d , FD and a finite sum over surviving words show that the natural density of $\mathbb{N} \setminus \mathcal{C}_d$ is their number divided by 2^d . It is at most $\frac{1}{2}\vartheta^{d-1}$ for sufficiently large d . Since $\mathcal{C}_d \subseteq \mathcal{C}_\infty$,

$$\overline{\text{dens}}(\mathbb{N} \setminus \mathcal{C}_\infty) \leq \frac{1}{2}\vartheta^{d-1}.$$

Letting $d \rightarrow \infty$ proves the assertion. In this argument $N \rightarrow \infty$ is taken first, with d fixed. $\square$

The conclusion concerns descent below the initial value. It does not prove that a descended value reaches 1, or that subsequent orbit samples avoid an exceptional set. Even a density-zero exceptional set can contain a nontrivial cycle. No interchange of the two limits, and no assertion of finite stopping time for every start, follows from Theorem 6.1.

7. Exact floor defects and the unresolved analytic estimates

For $n \geq 3$, write

$$X = n^{3/2}, \quad m = \lfloor X \rfloor, \quad \theta = X - m, \quad Y = m^{3/2}, \quad v = \lfloor Y \rfloor, \quad \theta_2 = Y - v.$$

Lemma 7.1 (one-signed linearization). One has

$$m^{3/2} = \frac{3}{2}mn^{3/4} - \frac{1}{2}n^{9/4} + E(n), \quad 0 \leq E(n) \leq \frac{3}{8}(n^{3/2} - 1)^{-1/2} \leq \frac{1}{2}n^{-3/4}.$$

Proof. Let $a = \sqrt{m}$, $b = \sqrt{X}$. Direct factorization gives

$$E = a^3 - \frac{3}{2}a^2b + \frac{1}{2}b^3 = \frac{1}{2}(a-b)^2(2a+b) \geq 0.$$

Moreover $\theta = b^2 - a^2$, and $Ea \leq \frac{3}{8}\theta^2$ follows from $4a(2a+b) \leq 3(a+b)^2$, equivalent to $(5a+3b)(a-b) \leq 0$. Thus $E \leq \frac{3}{8}m^{-1/2}$, which is at most $\frac{3}{8}(X-1)^{-1/2}$. Finally $(X-1)^{-1/2} \leq \frac{4}{3}X^{-1/2}$ for $n \geq 3$. $\square$

Lemma 7.2 (gap and carry). For an integer $h \geq 1$, set $\Delta_h X(n) = X(n + 2h) - X(n)$. Then

$$m(n + 2h) - m(n) = \lfloor \Delta_h X(n) \rfloor + \kappa_h(n), \quad \kappa_h(n) = \mathbf{1}_{\{X(n)\} + \{\Delta_h X(n)\} \geq 1} \in \{0, 1\}.$$

Proof. Expand $X(n) + \Delta_h X(n)$ into integer and fractional parts, and take its floor. The sum of the two fractional parts lies in $[0, 2)$. $\square$

For $1 \leq h \leq P/4$ and $n \in (P, 2P]$, differentiation gives $(\Delta_h X)'(n) \asymp hP^{-1/2}$. The level sets of $\lfloor \Delta_h X \rfloor$ consequently form $O(1 + hP^{1/2})$ intervals. Full intervals have length $\asymp P^{1/2}/h$; the two endpoint intervals can be shorter. A coefficient involving this floor is frozen only on these intervals. A derivative estimate proved there must be summed over the partition, or replaced by a separate global argument with controlled errors.

Lemma 7.3 (next-level defect). One has

$$v^{3/2} = m^{9/4} - \frac{3}{2}v^{1/2}\theta_2 - \mathcal{R}, \quad 0 \leq \mathcal{R} \leq \frac{3}{8}v^{-1/2}\theta_2^2.$$

Proof. Apply Taylor's theorem to $x^{3/2}$ between v and $v + \theta_2 = Y$. Its second derivative is $\frac{3}{4}x^{-1/2} \leq \frac{3}{4}v^{-1/2}$ on that interval, and is positive. Rearranging the Taylor formula gives the result. $\square$

These identities describe the objects requiring cancellation; they do not supply that cancellation. Two targets from the earlier draft are:

$$\left| \sum_{\substack{P < n \leq 2P \\ n \text{ odd}}} e\left(\frac{i}{2}n^{3/2} + \frac{j}{2}m^{3/2}\right) \right| \ll_{\varepsilon} P^{23/24+\varepsilon}, \quad |i| \leq 2P^{1/24}, \quad 1 \leq |j| \leq 2P^{1/24}, \quad (7.1)$$

and, with $c_k(n) = \frac{3k}{4}n^{9/8}$,

$$K_k(P) = \sum_{\substack{P < n \leq 2P \\ n \text{ odd}}} e(c_k(n)\theta_2(n)), \quad |K_k(P)| \ll_{\varepsilon} P^{95/96+\varepsilon}, \quad 1 \leq k \leq P^{1/24}. \quad (7.2)$$

The indices i, j, k are integers. Both estimates, with the printed uniformities, are **open targets in this note**. Equations (7.1) and (7.2) are not invoked in any unconditional proof above. Even if (7.2) were established, its transfer to the full mixed-mode families in Proposition 4.2 would still require proof. In particular, a coefficient $j/2$ can be a half-integer, so a lemma restricted to integer frequencies cannot be used without extending its hypotheses.

Short-interval variants require separate estimates. A bound for a transition set $\Omega \subset (P, 2P]$ of the form $|\Omega| \ll P^a$ implies only $|\Omega \cap I| \leq \min(|I|, O(P^a))$. It does not imply a bound proportional to $|I|/P$. Therefore the earlier localized-kernel threshold $|I| \geq P^{29/48+\delta}$ is not asserted here.

7.1 An unconditional estimate averaged over a shift

Proposition 7.4 (shift average). Let $L \geq 1$, let $A_1 < \dots < A_L$ obey $|A_t - A_s| \geq a|t - s|$ with $a > 0$, and let $x_1, \dots, x_L$ be real. For $S_{\lambda} = \sum_{t=1}^L e(A_t\{x_t + \lambda\})$,

$$\left| \int_0^1 |S_{\lambda}|^2 d\lambda - L \right| \leq \frac{4}{\pi} \frac{L}{a} (1 + \log L). \quad (7.3)$$

For every $0 < \eta < 1$, outside a set of shifts of measure at most η ,

$$|S_{\lambda}| \leq \left[\frac{L}{\eta} \left(1 + \frac{4}{\pi a} (1 + \log L) \right) \right]^{1/2}.$$

Proof. Expanding the square gives the diagonal term L . For $t \neq s$, the integrand $e(A_t\{x_t + \lambda\} - A_s\{x_s + \lambda\})$ is periodic in λ . Integrate over a period starting at one of its two fractional-part breakpoints. There is at most one other interior breakpoint, giving at most two intervals on which the phase is affine with slope $A_t - A_s$. On each interval the integral has modulus at most $1/(\pi|A_t - A_s|)$. Hence the off-diagonal contribution is bounded by

$$\frac{2}{\pi a} \sum_{t \neq s} \frac{1}{|t - s|} = \frac{4}{\pi a} \sum_{r=1}^{L-1} \frac{L - r}{r} \leq \frac{4}{\pi} \frac{L}{a} (1 + \log L).$$

For $L = 1$ the off-diagonal sum is empty. Markov's inequality applied to $|S_\lambda|^2$ gives the second assertion. $\square$

The estimate allows a small exceptional set of shifts. It gives no bound for a specified shift, including $\lambda = 0$ in (7.2).

8. Evidence, availability, and remaining work

The status of the principal statements is as follows.

Statement	Warrant in this version
Envelope, branch identity, minimal words, floor defects	Exact proofs in Sections 2, 5, and 7
Certificate density 3/4	Classical analytic proof in Theorem 3.1
Correlation-to-count and Fourier-to-parity implications	Proofs in Propositions 4.1 and 4.2
Certificate densities 13/16, 7/8	Conditional theorems; nested hypotheses unproved
Density-one finite certificates	Conditional on FD; no depth uniformity assumed
Deterministic nested sums and short-interval kernels	Unproved; no numerical threshold certified
Shift-averaged estimate	Direct integral proof in Proposition 7.4

Supporting software is available in the Balanced Ternary Mathematical Laboratory repository, <https://github.com/sneakyweasel/btlab>. The accompanying source package contains this manuscript, its LaTeX build, and an exact-arithmetic validation script. That script checks the finite certificate list, the indicator identity on a finite census, and selected algebraic inequalities. Those checks support reproducibility; they are not proofs of asymptotic cancellation.

Earlier repository audits and Lean modules concern identities, inequalities, and arithmetic from the previous draft. This version claims no Lean formalization of its analytic hypotheses or of the complete conditional manuscript. Earlier theorem numbers are superseded and should not be used as citations to unconditional nested estimates. Companion texts that rely on those estimates require their own review.

To obtain an unconditional improvement beyond the certificate density 3/4, it would suffice to establish the correlations identified in Theorem 5.2. To reach 7/8 by this route requires the additional five-letter correlations in Theorem 5.3. A proof of all fixed-depth counts would imply density-one certificates by Theorem 6.1, but a further argument would still be required to settle universal termination.

Acknowledgments and AI assistance

Large language models were used throughout the development of the earlier draft and this revision, including the formulation and review of proof arguments, drafting, programming, and discussion of Lean formalizations. The September 2026 review with OpenAI Codex identified unresolved analytic steps and prepared this conditional formulation, its proofs, and its release materials. AI assistance and automated checks are not independent mathematical validation. The author is responsible for the statements, proofs, code, and final approval of the preprint. The models are not authors.

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